

**SPhO 2017 Questions**

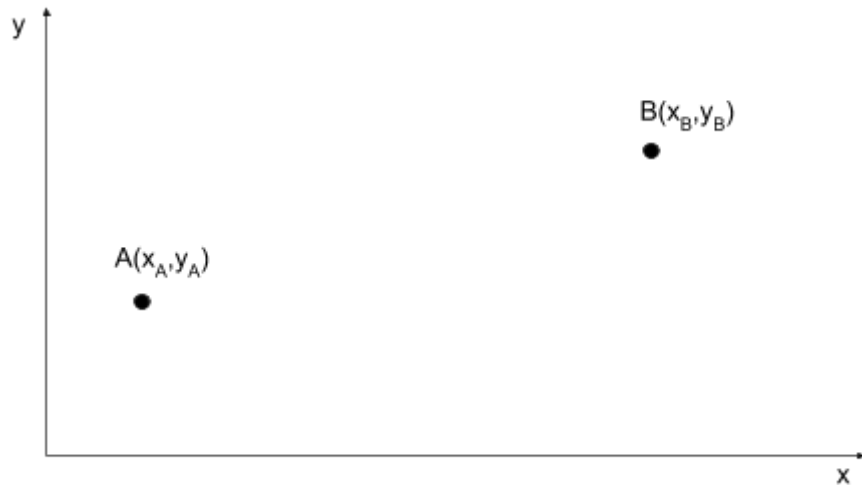
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**Part A (1.5h)**

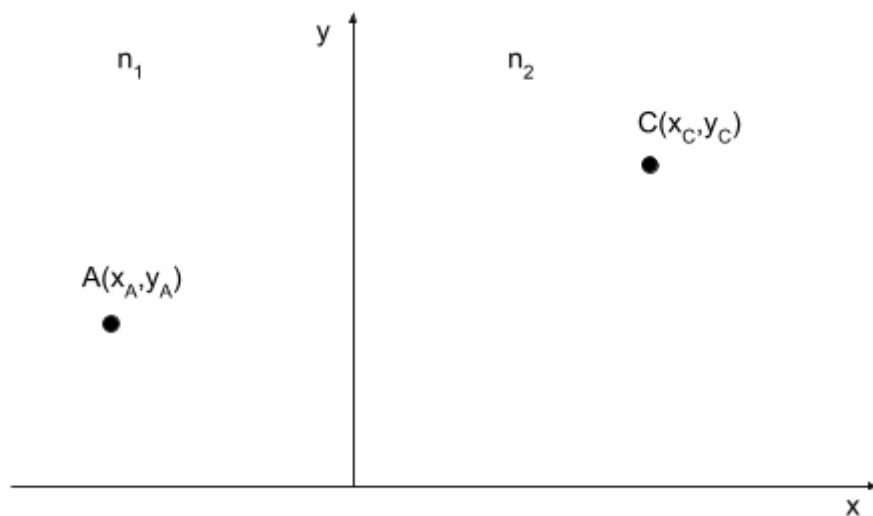
- 1) A planet is in a circular orbit around a star of mass  $M$ . The star is much more massive than the planet. The star explodes symmetrically and instantaneously such that its outer envelope is pushed far beyond the orbit of the planet (and does not interfere with the planet). The remaining piece of the star is of mass  $M'$  (still much greater than the mass of the planet). Find the eccentricity of the new orbit of the planet.

- 2) Fermat's principle states that light travels along the path of least time. The time taken to travel a unit distance in a medium is inversely proportional to the medium's refractive index,  $n$ .

- a) In the diagram below, draw the path of light from point  $A(x_A, y_A)$  to  $B(x_B, y_B)$ . Label it (a).

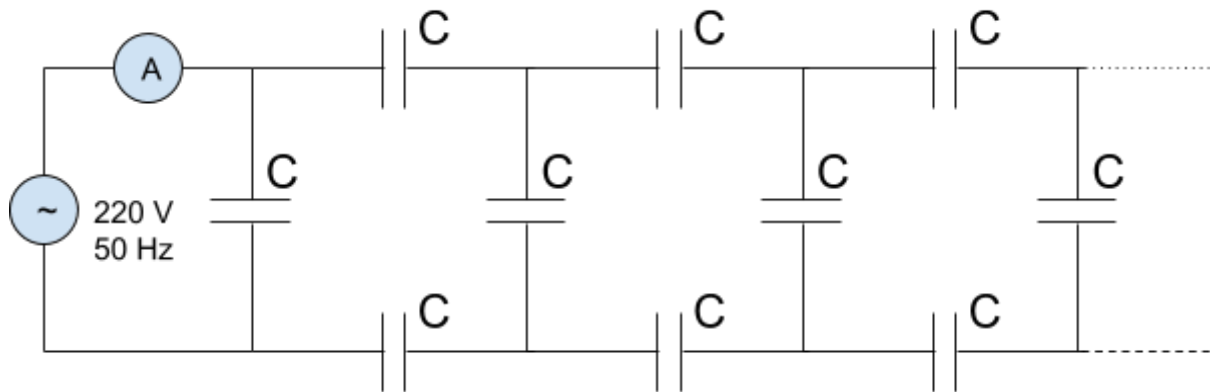


- b) The light ray now has to meet the x-axis at some point  $(x, 0)$  on its path while travelling from A to B.
- Express the distance travelled by the light ray from A to B.
  - Find and draw the path of light from point A to B on the same diagram above, and label it (b). Comment on the angles.
- c) In the diagram below, point A is to the left of the y-axis (refractive index  $n_1$ ) and point C is to the right of the y-axis (refractive index  $n_2$ ). Find the path of light from point  $A(x_A, y_A)$  to  $C(x_C, y_C)$ . Comment on the angles.



- 3)  $^{36}\text{Cl}$  decays into  $^{36}\text{Ar}$  (with 98.1% probability) and  $^{36}\text{S}$  (with 1.9% probability). A sample of water from an underground cave was found to contain  $20\mu\text{g}$  of  $^{36}\text{Cl}$  and  $0.36\mu\text{g}$  of  $^{36}\text{S}$ . Physicists then determined the age of the cave to be 290 000 years old. Find the half-life of  $^{36}\text{Cl}$ .

- 4) The infinite circuit below is powered by a 220V, 50Hz ideal AC voltage source. Suppose  $C = 1\mu\text{F}$ . What is the current as measured by the ideal AC ammeter?



**Part B (1.5h)**

- 5) A plane wave is travelling in space; frame  $S'$  is moving along the positive  $x$ -axis at constant velocity with respect to frame  $S$ . The Lorentz transformation is given by:

$$ct' = \gamma(ct - \beta x)$$

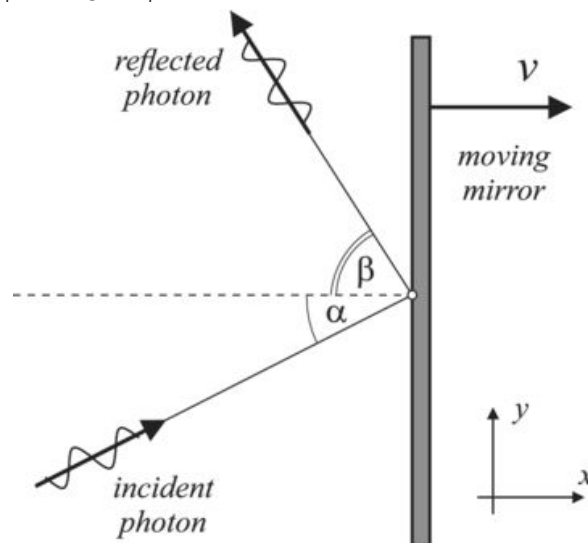
$$x' = \gamma(x - \beta ct)$$

$$y' = y$$

$$z' = z$$

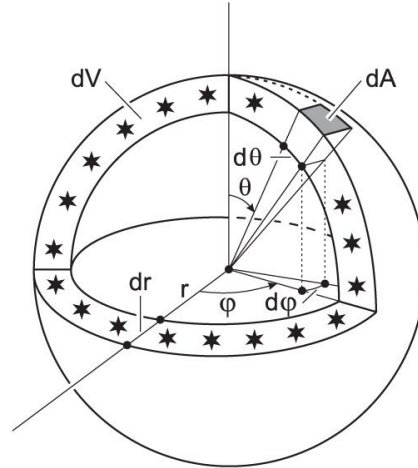
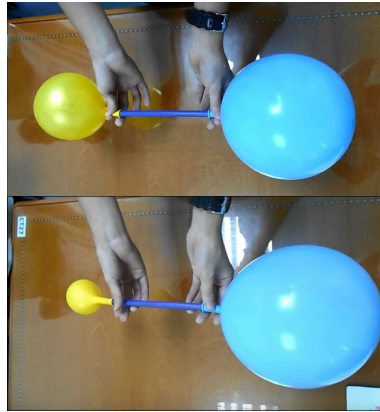
- a) For a plane wave, the quantity  $\mathbf{k} \cdot \mathbf{x} - \omega t$  is invariant with respect to relativistic transformations, i.e.  $\mathbf{k}' \cdot \mathbf{x}' - \omega' t' = \mathbf{k} \cdot \mathbf{x} - \omega t$ . Here,  $\mathbf{k}$  is the wave vector with magnitude  $k = |\mathbf{k}| = \omega/c$ . Express  $\omega'$  and  $ck'$  in terms of  $\mathbf{k}$ ,  $\omega$  and  $\gamma$ .

- b) A light ray is now incident on a moving mirror. (Note that Snell's law of reflection only applies in the stationary frame!) The light approaches the mirror with angular frequency  $\omega_i$  at angle  $\theta_i$  to normal; it leaves with angular frequency  $\omega_r$  at angle  $\theta_r$  to normal.



- i) Express  $\omega_r$  in terms of  $\theta_i$ ,  $\omega_i$  and  $\beta$  (which is  $v/c$ ).
- ii) What is the energy of a photon in the reflected ray?

- 6) In the following experiment, two balloons of different sizes are connected by a valve of cross-sectional area  $A$  (initially shut). Surprisingly, when the valve is opened, the smaller balloon shrinks while the bigger balloon expands.



The left figure shows an example set-up: 2 balloons connected by a valve. The smaller balloon (left) shrinks when the valve is opened while the larger balloon (right) expands. The right figure shows a balloon fragment, a small polar rectangle on the surface of the sphere. One pair of sides is the arc of angle  $d\theta$  of a circle of radius  $r$ ; the other pair is the arc of angle  $d\phi$  of a circle of radius  $r \sin\theta$ .

- a) The tension on the surface of the balloon is given by:  $F_r \propto \frac{r}{r_0^2} [1 - (\frac{r_0}{r})^6]$ .
- Show that the pressure in the balloon is given by  $P = \frac{C}{r_0^2 r} [1 - (\frac{r_0}{r})^6]$  for a constant  $C$ .
  - Show that the radius of the balloon at maximum pressure is given by  $r_{max} = 7^{1/6} r_0$ .
  - Plot the pressure in the balloon as a function of its radius.
- b) The smaller balloon initially has radius  $r_{si} = 7^{1/6} r_0$ . After the valve is opened, it shrinks to radius  $r_{sf} = 6^{1/6} r_0$  at equilibrium.
- Find the radius of the bigger balloon  $r_{bf}$  at equilibrium, to 3 significant figures in terms of  $r_0$ . Numerical methods are allowed.
  - Find the initial radius  $r_{bi}$  of the bigger balloon, to 3 significant figures in terms of  $r_0$ . Numerical methods are allowed.
  - Show that at constant temperature  $T$ , the time taken for the balloons to reach equilibrium after the valve is opened is given by:

$$t = \frac{16\pi}{3A} \sqrt{\frac{m}{kT}} \int_{r_{bi}}^{r_{bf}} \frac{r_b [1 + 2(\frac{r_0}{r_b})^6]}{\frac{1}{r_s} [1 - (\frac{r_0}{r_s})^6] - \frac{1}{r_b} [1 - (\frac{r_0}{r_b})^6]} dr_b$$

where  $m$  is the mass of the gas particles and  $k$  is Boltzmann's constant.

7) This question is regarding electrostatics.

- a) There is a volume  $V$  bounded by a surface  $S$ . Let the electric field outside the volume  $V$  as a function of position  $\mathbf{r}$  be  $E_{outside}^{\perp}(\mathbf{r})$  and the electric field inside the volume  $V$  be  $E_{inside}^{\perp}(\mathbf{r})$ . Show that  $E_{outside}^{\perp}(\mathbf{r}) - E_{inside}^{\perp}(\mathbf{r}) = \frac{\sigma(\mathbf{r})}{\epsilon_0}$ , where  $\sigma(\mathbf{r})$  is the surface charge density.
- b) Suppose  $V$  is a sphere with constant volume charge density  $\rho$ . Show that the charge on its surface  $S$  is everywhere 0.
- c) Now suppose  $V$  is a conducting sphere with charge  $Q$  and radius  $R$ . Find the force experienced by the right half of the sphere.